The Repair Bypasses the Gate: Context Noise Exposes the Mechanism of Own-Segment Bias in Active-Dendrite Continual Learning

Ranxi Zhou[a]

[a]*Hunan University of Chinese Medicine, Changsha, 410208, China*

Abstract

Active-dendrite continual-learning systems condition each neuron's plasticity and routing on a task-context signal. A companion audit ([1], S1) found that such networks never develop segment–task specialization and that a constant bias b on the own-segment dendritic input causally repairs the learning term ($\Delta\ell$ up to +4.6, $t \geq 8$). Every result in that line – including the repair itself – was obtained under oracle context. We ask the question the line left open: *when the context signal is wrong, what survives?* Under the worst-case evaluation protocol (each sample's context one-hot flipped to a uniformly random other task with probability ρ), the answer is: everything, at every noise level. Final accuracy moves by less than 0.05 points at $\rho \leq 0.4$ for both the base system and the repaired one; the repair's paired gain is retained at 94–101% (+1.22 → +1.23 on MNIST, +0.45 → +0.42 on Fashion-MNIST). Zero noise sensitivity is not robustness — it is a symptom. Mechanism probes show why: the trained dendritic gate is *context-degenerate*. Swapping the context one-hot changes at most 6.2% of predictions in the base network, and after the repair (b=10) exactly 0.0% – the bias saturates the winning segment's sigmoid so the gate output no longer depends on the context at all, while the dendritic

weight columns stay mutually orthogonal (cosine ≈ 0.003, i.e. their random initialization) throughout training. A degenerate-control ablation then asks what the repair actually repairs: a network with the entire dendritic gate removed ($h = \mathrm{kwta}(t)$, same trunk, same anchor chain) matches the repaired network on MNIST (73.22 vs. 72.63, n.s.) — there the repair's entire benefit is reproduced by gate removal, i.e. *the repair works by removing the dependence on the thing it was supposed to fix*. On Fashion-MNIST the bypass is rejected (66.84 vs. 64.30, t=6.8): the multiplicative gate retains input-dependent value even while context-degenerate. We conclude that "context-gated protection" in this architecture family runs, at these training budgets, on a gate that does not gate by context; noise-robustness certificates for such systems should therefore be read as *degeneracy diagnostics*, and repairs that survive adversarial context corruption deserve particular scrutiny.
Keywords: continual learning, active dendrites, context gating, robustness evaluation, measurement validity, catastrophic forgetting

1. Introduction

Active-dendrite architectures give each neuron a bank of dendritic segments and let a task-context signal select which segment gates the neuron's output [2, 3, 4]. In continual learning the arrangement is attractive for a specific reason: if each task's context activates its own segments, protection can be localized to exactly the parameters that task used, and interference across tasks should shrink. A sequence of audits from the same laboratory has stress-tested that promise. The original DeLP audit[1] corrected

[1] "Where do the gains come from?", [5]; the precursor study.

its own baselines and found most of the reported advantage was calibration, not mechanism. The companion dose–response audit (S1, [1]) found the specialization premise itself fails: segment–task routing never specializes, the standard routing instrument is contaminated by dead neurons, and yet a one-parameter intervention — a constant bias b on the own-segment dendritic input — causally improves the learning term in every cell of a $\{\text{dataset}\} \times \{\text{depth}\} \times \{\text{granularity}\}$ grid. A transfer study (S3, [6]) then showed the repair does not survive stronger backbones. One assumption has been shared by every experiment in this line, and by most of the active-dendrite continual-learning literature: *the context signal at evaluation is the correct one.*

This paper removes that assumption and reports what happens. The probe is deliberately adversarial: at evaluation time, each sample's context one-hot is flipped to a uniformly random *other* task with probability $\rho \in \{0.1, 0.2, 0.4\}$. Training is untouched. Three questions, preregistered before any data was collected (Sec. 3): (Q1) does the base gated system degrade? (Q2) does the S1 repair — whose premise is precisely that the gate should route by context — survive context corruption? (Q3) does a soft mixture mitigation help?

The empirical answers are monotone: nothing degrades, at any ρ, on either dataset, with or without the repair (Sec. 4). A robustness result this clean is not a reason to celebrate; it is a reason to dig. The rest of the paper is the dig. Mechanism probes (Sec. 5) show the trained gate is **context-degenerate**: its response to the context one-hot is statistically indistinguishable from its response to any other one-hot, and after the re-

pair the network's predictions are *exactly* invariant to context swaps. A degenerate-control ablation (Sec. 6) then locates what the repair actually does: on MNIST, deleting the dendritic gate outright reproduces the repair's full benefit — the repair works by *removing the system's dependence on the component it was supposed to be fixing.* On Fashion-MNIST the gate keeps input-dependent value that no context-free replacement provides, and the bypass interpretation is rejected.

The contribution is threefold:

1. **A negative-space result with a warning attached.** Adversarial context corruption is free at every noise level — because the mechanism it corrupts is not load-bearing at these training budgets. Any "robust to context noise" certificate earned by such a system is evidence of degeneracy, not of robustness.
2. **A mechanism re-qualification of the S1 repair.** The own-segment bias does not restore segment–task routing (which S1 showed never existed); it saturates the gate into a context-free multiplicative gain. On MNIST the repaired network is *equivalent to* a network with no dendritic gate at all; on Fashion-MNIST it is not, and the difference delineates what the dendritic bank is actually for.
3. **A reusable three-instrument protocol**: worst-case noise sweep → flip-rate/gate-response probe → degenerate-control ablation. Each instrument is cheap; together they convert a suspicious null into a mechanistic verdict. We preregistered the first two, and the third was forced on us by their result.

2. Background and setting

Architecture.. The reference network is the one shared by the S1/S3 line [1, 6]: a one-hidden-layer active-dendrite network (ADNet) with n_h=2000 hidden units, one dendritic segment bank per task (K=T), k-winners-take-all sparse output (k=64). Given input x and context one-hot c:

$$t = W_{\text{ff}}x, \qquad u_k = w_k^\top c \;\; (k\text{-th segment row}), \qquad h = \text{kwta}(t \odot \sigma(d)), \quad d = \max_k u_k, \tag{1}$$

i.e. the winning segment's scalar d gates every hidden unit multiplicatively. The repair of S1 adds a constant b to the own segment's u_{own} before the max, everywhere (train, eval, usage) [1].

Protection.. Usage-localized elastic consolidation (DeLP [5]): after each task, a Fisher anchor is stored and weighted by measured per-neuron usage; the penalty pulls shared parameters toward anchored values with strength λ=1000 (matching S1's dose–response setting). Segment-specific gradients are protected so each task trains its own dendritic block.

Protocol.. Permuted-MNIST and Permuted-Fashion-MNIST, T=10 tasks, 10 seeds, paired across all arms; permutations drawn once per seed and shared. Metrics: final average accuracy $\bar{a}$ over all tasks after the last task; paired t-statistics over seeds.

3. Preregistered design

The design was locked on 2026-10-25, before any data was written to disk (preregistration document included in the artifact). Noise is injected

at evaluation only; training sees oracle contexts. The noise model is worst-case in a specific sense: with probability ρ per sample the context one-hot is replaced by a uniformly random *other* task — never the correct one — so corrupted samples exercise the gate under a plausible wrong context rather than a neutral one.

- **E1 (gating robustness).** b=0, $\rho \in \{0, 0.1, 0.2, 0.4\}$, both datasets. Survival: ρ=0.1 costs $\leq$ 2 points and ρ=0.2 costs $\leq$ 5; death: ρ=0.1 costs > 5 or ρ=0.2 collapses ($\bar{a} < 50$ or cost > 10).
- **E2 (repair survival).** Same grid crossed with $b \in \{0, 10\}$, paired per $(\rho, \text{dataset}, \text{seed})$. Survival: at ρ=0.1 the repair's paired $\Delta\bar{a} > 0$ and retains $\geq 50\%$ of its ρ=0 gain; death: $\Delta\bar{a} \leq 0$.
- **E3 (soft mixture, exploratory).** Context replaced by $(1-\rho)$ onehot+ ρ uniform; direction only, no survival claim (multiple-comparison hygiene).

The noise stream is fixed per seed (a dedicated generator), so paired comparisons share corruption patterns exactly. 40 runs for E1+E2 (2 datasets $\times$ 2 arms $\times$ 10 seeds; each run evaluates all four ρ levels), 20 for E3.

4. Results: nothing degrades, at any noise level

Table 1 is the entire empirical content of E1 and E2. Both preregistered survival criteria pass, with margins that make the criteria look quaint: the largest movement anywhere is 0.03 points (Fashion base, ρ=0.4, and it is *upward*), the largest degradation is 0.01 points (MNIST base, ρ=0.4), where

Table 1: Final average accuracy under worst-case context flips (n=10 seeds; noise applied to the final-phase evaluation only).

dataset	arm	ρ=0	ρ=0.1	ρ=0.2	ρ=0.4
MNIST	base (b=0)	71.41	71.41	71.42	71.40
MNIST	repaired (b=10)	72.63	72.63	72.63	72.63
Fashion	base (b=0)	66.38	66.39	66.40	66.41
Fashion	repaired (b=10)	66.84	66.84	66.84	66.84

the criterion allowed 10. The repair's paired gain is +1.22 (t=3.8) at ρ=0 and +1.23 (t=3.8) at ρ=0.4 on MNIST; +0.45 and +0.42 on Fashion-MNIST (t=0.6, n.s. throughout — consistent with S3's finding that Fashion gains from the repair are small at T=10 [6]). Retention of the repair's gain at ρ=0.4 is 100.7% on MNIST and 93.8% on Fashion-MNIST (at ρ=0.1: 100.5% and 97.6%). E3's soft mixture moves nothing ($|\Delta| \leq 0.02$, all n.s.), which will make sense shortly.

A protocol-level artifact is worth excluding explicitly: the noise *is* reaching the network. Per-seed corruption patterns are fixed, the context tensor is modified before the forward pass, and — the direct check — swapping contexts *does* change predictions in the base network (6.2% of samples on average, up to 17.7% for particular swaps; Sec. 5). What is true is that prediction changes of this kind average out across a 10-way task mixture at the final-accuracy level. The null is real, and it demands a mechanism.

5. Mechanism: the gate does not gate by context

Three probes, each applied to fully trained networks. The headline numbers below are MNIST seed 0; the pattern replicates on Fashion-MNIST seed 0 and MNIST seed 1 (all numbers in the artifact): flip rates 6.2–6.9% mean / 14.9–19.0% worst at b=0, exactly 0.0 at b=10, gate own/other identical to four decimals, and mean column cosines in $[0.001, 0.003]$ in every case.

Prediction flip rate.. For each task c and each alternative context $c' \neq c$, measure the fraction of test samples whose argmax prediction differs between ctx=c and ctx=c'. Base network: mean 6.2%, worst pair 17.7% — the network's output is *mostly* context-invariant already. Repaired network (b=10): mean and worst **exactly** 0.0%. After the repair there is no context pair, anywhere in the task grid, that changes a single prediction.

Gate response.. The winning segment's gate value $\sigma(d)$ has mean 0.5649 under the correct context and 0.5649 under every alternative — identical to four decimal places, in both the base and repaired networks. The context one-hot selects *which* segment wins, but not *how much* it gates.

Column geometry.. The dendritic weights projected onto the T context columns remain mutually orthogonal after training: mean pairwise cosine 0.003 (per-model mean in $[0.001, 0.003]$; the most negative single pair over the three models is -0.055), indistinguishable from their random initialization. Training, with per-task gradient protection, leaves the context channel of the gate bank essentially at its prior.

Together: the architecture's context pathway is present, trainable in principle, protected per task — and unused. The max over segments is won by

Table 2: Degenerate control vs. the repair (n=10 paired seeds, ρ=0).

dataset	repaired (b=10)	no-gate control	base (b=0)	no-gate − base
MNIST	72.63	**73.22**	71.41	+1.81*
Fashion	**66.84**	64.30	66.38	−2.09*

whichever row the (orthogonal, near-random) columns happen to favor for the current input batch statistics, and the repair adds a constant so large (b=10 against $|u| \lesssim 1$) that the max is always won by the own segment, with $\sigma(10 + u) \approx 1$ saturated: a context-free gain of one. Equation 1 degenerates to $h = \mathrm{kwta}(t)$.

6. Ablation: what does the repair repair?

If the repaired network computes (approximately) $h = \mathrm{kwta}(t)$, then a network with the dendritic bank deleted and the same trunk, sparsifier, and anchor chain should reproduce the repair's benefit. We built that degenerate control twice — the first version was invalid (it applied an extra sigmoid to t, changing the activation structure; the mistake is retained in the artifact as a documented correction) — and report the corrected one: $h = \mathrm{kwta}(W_{\mathrm{ff}}x)$, plain Fisher anchors on the trunk and readout, everything else identical.

MNIST: bypass confirmed. The no-gate control (73.22) matches the repaired network (72.63; paired difference −0.59, t= − 1.0, n.s.) and beats the base (+1.81, t=3.4). Everything the repair buys on MNIST is bought by removing the gate. The repair's causal pathway and the gate's deletion have the same signature because they are the same operation: the bias implements a soft, saturating deletion of the dendritic modulation.

Fashion: bypass rejected — and informative. The repaired network (66.84) beats the no-gate control (64.30) by +2.54 (t=6.8). Here the multiplicative gate contributes value that no context-free trunk can, even though the gate is context-degenerate (Sec. 5): a per-unit, input-dependent scaling $t \odot \sigma(d(x))$ is a capacity the plain trunk lacks. The S3 transfer study [6] asked whether the repair crosses architectures and answered no; this ablation asks a sharper question — *what is the gate for, once the repair exists?* — and gets a dataset-split answer: on MNIST, nothing; on Fashion, a learnable per-unit gain.

One confound is acknowledged rather than hidden: the dendritic arms use usage-weighted Fisher anchors (DeLP) while the no-gate control uses plain Fisher anchors, both at λ=1000; usage weighting reduces effective anchor strength, so the comparison slightly *favors* the no-gate control. The MNIST equivalence and the Fashion gap are therefore lower and upper bounds respectively of the gate's contribution, which is the right direction for both conclusions.

7. Discussion

For the S1 story.. The dose–response audit established that the own-segment bias repairs the learning term; the mechanism section of that paper showed the network never pushes back at the parameter level. This paper supplies the missing re-qualification: the repair does not restore routing — there was no routing to restore — it *saturates the gate into a context-free gain*. Read together, the chain is now: no specialization (S1) → repair works (S1) → repair works without the gate (this paper, MNIST) → but the gate still pays

on harder data (this paper, Fashion) → the repair does not transfer to strong backbones (S3). What survives as the "mechanism" of the repair is sparse capacity management, not context-gated routing.

For evaluation practice.. Robustness certificates obtained under context corruption are usually read as good news. In this family they are a diagnostic: a system whose accuracy is invariant to *adversarial* context corruption is a system whose context pathway is not load-bearing. The three-instrument protocol (Sec. 4–6) costs a few GPU-hours and converts that ambiguity into a verdict. We suggest it as a standing companion to any context-gated continual-learning evaluation, in the same spirit as the dead-neuron probe of S1.

Limitations.. (i) Noise is injected at evaluation only; training-time context errors corrupt usage statistics and anchors differently, and are the natural next experiment. (ii) The mechanism probes are computed on fully trained networks at T=10; whether the context columns stay orthogonal at larger T or longer training is untested. (iii) The degenerate control differs from the dendritic arms in anchor weighting (usage vs. plain), as discussed above. (iv) One architecture family, two digit datasets — the same scope limits as the companion line.

8. Conclusion

Asked whether context-gated protection survives a wrong context signal, the system answered: it never needed the signal. The repair that was motivated by broken routing survives for the same reason — it never used the

routing. Only on Fashion-MNIST, where the gate's multiplicative structure carries input-dependent value, does something context-irreplaceable remain. The general lesson travels beyond this architecture: *when a robustness probe returns exactly zero damage, measure the pathway before celebrating the armor.*

Data and code availability

All code, per-run records (including per-ρ accuracy vectors), the preregistration document, and both versions of the degenerate control (including the invalidated first attempt) are in the artifact repository https://github.com/zhouranxi/delp-anchor-budget (folder `S4_gate_bypass/`, self-contained: paper sources, scripts, raw records), alongside the companion papers S1–S3.

Declaration of competing interest

The author declares no competing interests.

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